

In single-variable calculus, when our function $f(x)$ was defined over a closed interval $[a, b]$, we also investigated finding *global* extrema. In particular, these values can occur at any local min/max or at the *endpoints* of the interval, a and b . What does this mean in the multivariable context? Well, global extrema can now occur at any points on the boundary of the domain. This poses a serious problem since we cannot simply ‘evaluate’ a function $z = f(x, y)$ at the boundary—a set containing infinitely many points. A sample problem might be:

Example. Find the global extrema of $f(x, y) = 2 + 2x + 4y - x^2 - y^2$, where f is defined on the triangular region in the first quadrant of the xy -plane bounded by $x = 0$, $y = 0$, and $y = 9 - x$.

Specifically, we want to solve the following:

- *Objective:* Find the global extrema of $f(x, y) = 2 + 2x + 4y - x^2 - y^2$.
- *Constraint:* The triangular region in the first quadrant of the xy -plane bounded by $x = 0$, $y = 0$, and $y = 9 - x$ (i.e., a $z = 0$ level surface of some constraint function g).

So how do we streamline this difficult process? Fortunately, we have the following observation:

FACT: If we want to determine an extreme value of a multi-variable function f subject to a constraint represented by a level surface of a function g , then at the point which represents the solution, the graphs of the two functions must be tangent. This means that they share a tangent plane and that their gradients must be parallel. That is, there exists a constant $\lambda \in \mathbb{R}$ such that $\nabla f = \lambda \nabla g$.

Note: This is only true when considering the *boundary* of a planar domain. When comparing to the interior extrema, you must also use the Hessian and the techniques from Section 13.7.

Theorem.[Lagrange Multipliers] Suppose that $f(x, y, z)$ and $g(x, y, z)$ are differentiable and $\nabla g \neq \vec{0}$ when $g(x, y, z) = 0$. To find the global extrema of f subject to the constraint $g(x, y, z) = 0$ (if these exist), find the values of x, y, z , and λ that simultaneously solve the following system:

$$\begin{cases} \nabla f = \lambda \nabla g \\ g(x, y, z) = 0 \end{cases}$$

To find the global extrema of f , evaluate f at each of these points. The largest value is the maximum of f and the smallest value is the minimum value of f (subject to the constraint g).

Example. Find the global extrema of $f(x, y) = xy$ if it is defined on $\frac{x^2}{8} + \frac{y^2}{2} = 1$.

Example. Find the global extrema of $f(x, y) = 3x + 4y$ if it is defined on $x^2 + y^2 = 1$.

Sometimes we want to impose multiple constraints. To handle this, we just introduce a new multiplier! Traditionally, this second multiplier is given by the constant μ .

Theorem.[Lagrange Multipliers with Two Constraints] Suppose that $f(x, y, z)$, $g_1(x, y, z)$, and $g_2(x, y, z)$ are differentiable, where ∇g_1 , and ∇g_2 are not parallel. To find the global extrema of f subject to the constraints

$$g_1(x, y, z) = 0 \quad \text{and} \quad g_2(x, y, z) = 0,$$

find the values of x, y, z, λ , and μ that simultaneously solve the following system:

$$\begin{cases} \nabla f = \lambda \nabla g_1 + \mu \nabla g_2 \\ g_1(x, y, z) = 0 \\ g_2(x, y, z) = 0 \end{cases}.$$

To find the global extrema of f , evaluate f at each of these points. The largest value is the maximum of f and the smallest value is the minimum value of f (subject to the constraints g_1 and g_2).

Example. The plane $x + y + z = 1$ cuts the cylinder $x^2 + y^2 = 1$ in an ellipse. Find the points on the ellipse that lie closest to and farthest from the origin.